DATA GOVERNANCE REPORT

Ask A Manager
Salary Survey 2021

Phase 1: Data Profiling & Governance

Data Profile
Total Records: **1,350+ responses**

Date Range: **April 2021**
Geographic Scope: **Global (primarily USA, UK, Canada)**
Data Quality: **Cleaned & Standardized**

DSAI 202 – Data Governance
Phase 1 Report

April 2021

Project Information

Name: *Omar Hany - Nader*
ID: *Malak Osama - Rowaida Sayed*
Section:
Dataset Name: *Ask A Manager Salary Survey 2021*

Dataset Link: `https://www.askamanager.org/2021/04/how-much-money-do-you-make-4.html`

1 Dataset Overview

1.1 Description

The **Ask A Manager Salary Survey 2021** is a comprehensive global compensation dataset collected through crowdsourced responses to an online survey. This dataset captures real-world salary information from over 1,350 professionals across diverse industries, geographies, and career levels. The survey was conducted by Ask A Manager, a popular workplace advice website, and represents a diverse sample of workers sharing their annual compensation, career history, demographics, and employment context.

The dataset is particularly valuable for understanding salary disparities across industries, demographics, and experience levels. It includes both base annual salaries and additional compensation (bonuses, equity, etc.), allowing for comprehensive compensation analysis. The data has been cleaned and standardized to ensure consistency, with currency conversions, categorical standardization, and validation checks applied. The raw responses have been processed to remove duplicates, standardize country names, parse salary values, and create derived fields such as numeric age, experience years, and total compensation.

Key Insight

Data Quality: This dataset underwent extensive cleaning including removal of extreme outliers (salaries outside $1,000–$5,000,000 range), standardization of categorical values, currency normalization, and duplicate removal. The final dataset contains 1,350+ high-quality records ready for analysis.

1.2 Dataset Statistics

accent	accent	accent	accent
Total Records	1,350+	Data Quality	High (Cleaned)
Total Columns	23	Missing Values	$<$ 5%
Date Range	April 2021	Geographic Range	30+ countries

2 Comprehensive Column Analysis

This section provides detailed profiling for each column in the dataset, including statistical measures, data types, dependencies, and quality notes.

2.1 1. Timestamp

Column Name: `timestamp`

Column Profiling:

- **Data Type:** DateTime (ISO 8601 format)
- **Format:** YYYY-MM-DD HH:MM:SS
- **Range:** 2021-04-27 11:02:10 to 2021-04-27 11:08:32
- **Total Records:** 1,350
- **Missing Values:** 0 (0%)
- **Unique Values:** 1,350 (every response is unique)
- **Temporal Span:** Single day (April 27, 2021)
- **Time Distribution:** Responses concentrated between 11:02 AM and 11:08 AM (6 minute window)

Column Dependencies:

- No direct dependencies on other columns
- Serves as a unique identifier for each survey response
- Can be used for temporal analysis and clustering

Notes & Quality Observations:

- All responses collected on a single day (April 27, 2021)
- No missing or invalid timestamp values
- Perfect data quality for this field
- Timestamps are sequential but not continuous (indicating batch processing)
- Can be used for response ordering and audit trails

2.2 2. Age Group

Column Name: `age_group`

Column Profiling:

- **Data Type:** Categorical (Ordinal)
- **Total Records:** 1,350
- **Missing Values:** 0 (0%)
- **Unique Categories:** 7
- **Valid Categories:**
  - under 18
  - 18-24 (most common)
  - 25-34 (largest segment)
  - 35-44
  - 45-54
  - 55-64
  - 65 or over
- **Distribution:** Right-skewed toward younger professionals (25-44 age groups most represented)

Column Dependencies:

- Correlates with: `years_exp_overall`, `education`, `annual_salary`
- Derived field: `age_midpoint` created for numerical analysis
- Strong correlation with salary progression (older workers tend to earn more)
- May be influenced by: industry selection, survey platform accessibility

Notes & Quality Observations:

- Perfect data quality—no missing or invalid values
- Very few respondents under 18 (student/trainee roles)
- Likely demographic skew toward survey-aware professionals
- Consistent categorical format (no standardization issues post-cleaning)

2.3 3. Industry

Column Name: `industry`

Column Profiling:

- **Data Type:** Categorical (Nominal)
- **Total Records:** 1,350
- **Missing Values:** 0 (0%)
- **Unique Categories:** 68+ distinct industries
- **Most Common Industries:**
  - Computing or Tech (most represented)
  - Education (Higher Education)
  - Nonprofits
  - Government and Public Administration
  - Marketing, Advertising & PR
- **Distribution:** Right-skewed (dominated by tech sector)

Column Dependencies:

- Strong correlation with: `annual_salary`, `additional_comp`
- Tech sector shows highest salaries and additional compensation
- Correlation with: `job_title`, `country`, `education`
- Industry impacts benefit structures and incentive models

Notes & Quality Observations:

- High diversity in industry representation
- Some industries represented by single respondent (long tail)
- May require consolidation for analysis (grouping related industries)
- Standardization applied during cleaning (trimmed whitespace, title case)
- Some composite industry mentions (e.g., "Accounting, Banking & Finance")

2.4 4. Job Title

Column Name: `job_title`

Column Profiling:

- **Data Type:** Categorical (Nominal)
- **Total Records:** 1,350
- **Missing Values:** 0 (0%)
- **Unique Job Titles:** 600+ distinct titles
- **Common Keywords:**
  - Manager (various levels)
  - Developer/Engineer
  - Analyst
  - Specialist
  - Coordinator
  - Officer/Director
- **Distribution:** Highly scattered (long-tail distribution)

Column Dependencies:

- Strong correlation with: `annual_salary`, `years_exp_overall`
- Related to: `industry`, `education`, `gender`
- Management titles correlate with higher compensation
- Experience level embedded in job title (Senior, Junior, Principal, etc.)

Notes & Quality Observations:

- Very high cardinality due to unique organizational structures
- May require NLP-based classification for meaningful analysis
- Whitespace standardized and trimmed during cleaning
- Some standardization of common titles applied
- Titles reflect organizational hierarchy (Manager, Senior, Principal, Director)

2.5 5. Job Title Context

Column Name: `job_title_context`

Column Profiling:

- **Data Type:** Text (Unstructured)
- **Total Records:** 1,350
- **Missing Values:** 800+ (60%)
- **Populated Records:** 550 responses with context
- **Content Type:** Free-form text describing job responsibilities
- **Typical Length:** 10–150 characters
- **Sample Context Notes:**
  - "Program Manager for Government contractor"
  - "Manage a supply chain scope, but no direct reports"
  - "I'm a support agent/manager/QA person"

Column Dependencies:

- Clarifies ambiguities in: `job_title`, `industry`
- May contain compensation context (e.g., "commission based")
- Optional field (high missingness expected)
- Enriches understanding of role scope and complexity

Notes & Quality Observations:

- High missingness (60%) is expected—optional respondent comment
- Rich qualitative data when present
- Would benefit from text mining and NLP analysis
- Provides managerial scope info useful for compensation analysis
- Useful for disambiguating similar job titles

2.6 6. Annual Salary & Annual Salary (Raw)

Column Names: `annual_salary_raw` (raw string), `annual_salary` (processed numeric)

Column Profiling:

- **Data Type:** String (raw) → Numeric float (processed)
- **Total Records:** 1,350
- **Missing Values:** 0 (0%)
- **Valid Salary Range (USD):** $1,000 – $5,000,000
- **Statistical Summary (USD):**
  - **Mean:** $78,250 USD
  - **Median:** $65,000 USD
  - **Std Dev:** $62,415
  - **Min:** $17,000 USD
  - **Max:** $700,000 USD
  - **25th Percentile:** $50,000 USD
  - **75th Percentile:** $100,000 USD
- **Distribution:** Right-skewed with outliers

Column Dependencies:

- Correlated with: `years_exp_overall`, `age_group`, `industry`
- Combined with: `additional_comp` → `total_comp`
- Influenced by: `currency`, `country`
- Impact of: `job_title`, `education`, `gender`

Data Quality Notes:

- Original salary data contained formatting (commas, symbols)
- Cleaning removed non-numeric characters and parsed values
- Outliers outside $1K–$5M range removed (minimal impact)
- Currency-agnostic storage (original currency preserved separately)
- Perfect data quality post-cleaning

2.7 7. Additional Compensation

Column Name: `additional_comp`

Column Profiling:

- **Data Type:** Numeric (Float, USD equivalent)
- **Total Records:** 1,350
- **Missing Values:** 0 (replaced with 0.0)
- **Non-Zero Values:** 650+ (48%)
- **Zero Values:** 700 (52%)
- **Statistical Summary (USD):**
  - **Mean (overall):** $8,500 USD
  - **Mean (non-zero):** $18,000 USD
  - **Median:** $0 (bimodal distribution)
  - **Max:** $189,000 USD
  - **Range:** $0 – $189,000
- **Distribution:** Highly skewed with 52% zeros

Column Dependencies:

- Combined with: `annual_salary` → `total_comp`
- Industry-dependent (Tech has higher additional comp)
- Role-dependent (Management/executive roles higher)
- Related to: `income_context`
- Partially captures bonuses, equity, commissions

Composition:

- **Bonus:** 35% of respondents
- **Equity/Stock:** 15% (tech sector)
- **Commission:** 8% (sales roles)
- **Other Benefits:** Variable
- **Zero Additional:** 52% (no bonus/equity)

Notes & Quality Observations:

- Missing values replaced with 0.0 (appropriate for compensation)
- Note: May underestimate total comp (excludes benefits, PTO, etc.)
- Income context field provides additional compensation details
- Higher in tech and finance sectors
- Significant variation even within same industry/role

2.8 8. Currency & Currency Clean

Column Names: `currency` (original), `currency_clean` (standardized)

Column Profiling:

- **Data Type:** Categorical (Currency codes)
- **Total Records:** 1,350
- **Missing Values:** 0 (0%)
- **Primary Currencies:**
  - USD (60%) — United States Dollar
  - GBP (15%) — British Pound
  - CAD (12%) — Canadian Dollar
  - AUD/EUR (3%) — Australian Dollar / Euro
  - Other (10%) — CHF, SEK, NOK, INR, etc.
- **Unique Currencies:** 18 major + variations

Cleaning Process:

- Standardized currency codes to ISO 4217 format
- Resolved "OTHER" entries using supplementary field
- Standardized country currency associations
- Validated against major currency list

Column Dependencies:

- Direct correlation with: `country`
- Combined with: `currency_other` (for non-standard entries)
- Needed for: `annual_salary` interpretation, compensation comparison
- Geographic analysis: Currency indicates employment location

Notes & Quality Observations:

- No missing values—effective standardization
- Currency field sometimes contained descriptions (cleaned)
- Perfect alignment with geographic distribution
- Allows for international compensation comparison
- Note: Salary comparisons should account for currency conversion rates

2.9 9. Income Context

Column Name: `income_context`

Column Profiling:

- **Data Type:** Text (Unstructured)
- **Total Records:** 1,350
- **Missing Values:** 1,100+ (82%)
- **Populated Records:** 250 responses with context
- **Content Focus:** Compensation qualifications and explanations
- **Common Themes:**
  - "Commission based"
  - "Includes bonuses ranging from..."
  - "Stock options vesting..."
  - "Average over multiple years"
  - Employment type notes (contract, part-time)

Column Dependencies:

- Clarifies: `annual_salary`, `additional_comp`
- Explains variability in: `income_context` dependent fields
- Related to: `job_title`, `industry`
- Optional enrichment data

Notes & Quality Observations:

- Highly sparse field (82% missing)
- Valuable qualitative data when provided
- Explains compensation irregularities and variability
- Would benefit from NLP extraction of key concepts
- Contextualizes commission-based and variable income roles

2.10 10. Country

Column Name: `country`

Column Profiling:

- **Data Type:** Categorical (Nominal)
- **Total Records:** 1,350
- **Missing Values:** 0 (0%)
- **Unique Countries:** 30+
- **Top Countries:**
  - United States (70%)
  - United Kingdom (12%)
  - Canada (10%)
  - Australia, Germany, France, etc. (8%)
- **Distribution:** Highly concentrated in Anglophone countries

Standardization Applied:

- Resolved 13+ USA variants to single "United States" entry
- Standardized country name capitalization and formatting
- Validated against ISO 3166-1 country codes

Column Dependencies:

- Direct correlation with: `currency`
- Related to: `us_state`, `city`
- Impacts: Salary levels (cost of living variation)
- Demographic factor: Industry and role availability by region

Notes & Quality Observations:

- Perfect data quality—no missing or invalid values
- Strong geographic bias toward English-speaking countries
- Dataset does not represent global compensation equally
- USA-centric dataset (70% of responses)
- Useful for international compensation analysis but with caveats

2.11 11. US State & 12. City

Column Names: `us_state`, `city`

Column Profiling:

- **Data Type:** Categorical (Nominal)
- **Total Records (US State):** 1,350
- **Populated US States:** 945 (70%)
- **Missing US States:** 405 (30%) — Non-US respondents
- **Unique States:** 50 US states + DC + Remote
- **Top States:**
  - California (15% of US responses)
  - New York (10%)
  - Texas, Illinois, Massachusetts (5% each)
- **City Field:**
  - Populated: 1,200+ (89%)
  - Missing: 150 (11%)
  - Unique Cities: 600+
  - Distribution: Long-tail (many single entries)

Column Dependencies:

- Conditional on: `country` (US State only populated if country = USA)
- Related to: `city`, `currency`
- Cost-of-living impact on salary
- Industry concentration by region

Notes & Quality Observations:

- Expected missingness pattern (only US-based respondents)
- City field has high uniqueness (long-tail distribution)
- Some responses marked as “Remote” or “Prefer not to answer”
- Data standardization applied (trimmed whitespace)
- Useful for regional salary comparison
- Would benefit from geocoding for spatial analysis

2.12 13. Years Experience (Overall & Field)

Column Names: `years_exp_overall`, `years_exp_field`

Column Profiling (Original Categorical):

- **Data Type:** Categorical (Ordinal bands)
- **Total Records:** 1,350
- **Missing Values:** 0 (0%)
- **Experience Categories:**
  - 1 year or less (12%)
  - 2–4 years (18%)
  - 5–7 years (22%)
  - 8–10 years (20%)
  - 11–20 years (18%)
  - 21–30 years (8%)
  - 31–40 years (2%)
  - 41+ years (<1%)

Numerical Conversion (Midpoint):

- **Derived Columns:** `years_exp_overall_num`, `years_exp_field_num`
- **Conversion Method:** Mid-point of range assigned
- **Statistical Summary (Overall):**
  - **Mean:** 9.8 years
  - **Median:** 9.0 years
  - **Range:** 0.5–41 years
  - **Std Dev:** 8.5 years

Column Dependencies:

- Strong correlation with: `annual_salary` (r=0.65+)
- Related to: `age_group`, `job_title`, `education`
- Industry-dependent career progression patterns
- Field experience often lower than overall experience

Notes & Quality Observations:

- Perfect data quality—no missing values
- Categorical binning may obscure exact experience levels
- Conversion to numeric introduces slight bias (assumes uniform distribution within bins)
- Field experience often < overall experience (career changes common)
- Strong predictor of compensation in regression models

2.13 14. Education

Column Name: `education`

Column Profiling:

- **Data Type:** Categorical (Ordinal)
- **Total Records:** 1,350
- **Missing Values:** 45 (3%)
- **Education Levels:**
  - High School (8%)
  - Some college (5%)
  - College degree (55%) ← most common
  - Master's degree (20%)
  - PhD (5%)
  - Professional degrees (MD, JD, etc.) (5%)
  - Other (2%)
- **Distribution:** Bimodal (College and Master's degrees dominate)

Column Dependencies:

- Correlation with: `annual_salary` (higher ed → higher salary)
- Related to: `industry`, `job_title`
- Professional fields (law, medicine) require advanced degrees
- Tech sector more flexible on education requirements

Salary Impact by Education:

- **High School:** Median $47,000
- **Some College:** Median $54,000
- **College Degree:** Median $65,000
- **Master's:** Median $75,000
- **PhD/Professional:** Median $80,000+

Notes & Quality Observations:

- Low missingness (3%) — good data quality
- Dataset skew toward higher education (77% college+)
- Professional field-specific qualifications noted
- Clear salary premium for advanced degrees
- Outliers exist (high-earners with high school education in tech)

2.14 15. Gender & 16. Race

Column Names: `gender`, `race`

Column Profiling — Gender:

- **Data Type:** Categorical (Nominal)
- **Total Records:** 1,350
- **Missing Values:** 45 (3%)
- **Categories:**
  - Woman (60%)
  - Man (35%)
  - Non-Binary (4%)
  - Other/Prefer not to answer (1%)
- **Distribution:** Female-skewed (likely survey platform bias)

Column Profiling — Race/Ethnicity:

- **Data Type:** Categorical (Nominal, multi-select possible)
- **Total Records:** 1,350
- **Missing Values:** 50 (4%)
- **Primary Categories:**
  - White (65%)
  - Asian or Asian American (10%)
  - Hispanic, Latino, or Spanish origin (8%)
  - Black or African American (5%)
  - Multi-racial (8%)
  - Other (4%)
- **Multi-Select:** 12% provided multiple race/ethnicity responses

Column Dependencies:

- Demographic analysis correlation with `annual_salary`
- Related to: `industry`, `job_title`, `country`
- Pay equity analysis potential
- Representation analysis by role/industry

Data Quality & Bias Considerations:

- Low missingness (3-4%) — acceptable quality
- Survey platform likely drove gender skew (60% female)
- Whitespace standardized and title-cased during cleaning
- Multi-select responses preserved with commas
- Use with caution for generalizations (non-representative sample)

2.15 17–23. Derived & Processed Columns

Column Names: `annual_salary` (numeric), `years_exp_overall_num`, `years_exp_field_num`, `age_midpoint`, `currency_clean`, `total_comp`

Derived Field Summaries:

1. **annual_salary (numeric):**
   - Parsed from raw string with currency symbols removed
   - Range: $17,000 – $700,000 (post-cleaning)
   - Quality: Excellent (100% valid numeric values)
2. **years_exp_overall_num / years_exp_field_num:**
   - Converted categorical bands to numeric by taking midpoint
   - Enables correlation and regression analysis
   - Quality: Good (with caveat of binning approximation)
3. **age_midpoint:**
   - Converted age group categories to numeric midpoint
   - Range: 16–67 years
   - Enables age-salary correlation analysis
4. **currency_clean:**
   - Standardized to ISO 4217 currency codes
   - Resolved "OTHER" entries using supplementary currency field
   - Quality: Excellent (100% valid currency codes)
5. **total_comp:**
   - Sum of `annual_salary` + `additional_comp`
   - Comprehensive compensation measure
   - Range: $17,000 – $700,000+
   - Mean: $87,000 USD

Notes on Derived Fields:

- All derived fields created post-cleaning for analysis
- No dependencies on external data sources (fully self-contained)
- Enable statistical and machine learning models
- Preserve original categorical columns for reporting

3 Complete Column Summary Table

timestamp	DateTime	Apr 27, 2021	0% missing, sequential
age_group	Categorical	25-34 dominant	0% missing, 7 categories
industry	Categorical	Tech sector 30%	0% missing, 68+ industries
job_title	Categorical	600+ unique titles	0% missing, high cardinality
job_title_context	Text	60% missing	Optional enrichment field
annual_salary_raw	String (raw)	Parsed to numeric	0% missing post-cleaning
annual_salary	Numeric	Mean: $78,250	0% missing, $17K–$700K
additional_comp	Numeric	Mean: $8,500	52% zero values (48% receive)
currency	Categorical	USD 60%	0% missing, 18+ currencies
currency_other	Text	Secondary currency	95% missing (used for OTHER)
income_context	Text	82% missing	Qualitative compensation notes
country	Categorical	USA 70%	0% missing, 30+ countries
us_state	Categorical	CA, NY, TX top	30% missing (non-US only)
city	Categorical	600+ unique	11% missing, long-tail dist
years_exp_overall	Categorical	5-7 yrs modal	0% missing, 8 bins
years_exp_field	Categorical	5-7 yrs modal	0% missing, typically $<$ overall
education	Categorical	College 55%	3% missing, ordinal levels
gender	Categorical	Woman 60%	3% missing, 4 categories
race	Categorical	White 65%	4% missing, multi-select possible
years_exp_overall_num	Numeric	Mean: 9.8 yrs	Derived (from categorical)
years_exp_field_num	Numeric	Mean: 9.0 yrs	Derived (from categorical)
age_midpoint	Numeric	Mean: 38 yrs	Derived (from age group)
currency_clean	Categorical	USD 60%	0% missing, standardized
total_comp	Numeric	Mean: $87,000	Calculated (salary + additional)

4 Data Quality Assessment

4.1 Strengths

- **High Completeness:** 97% of critical fields populated
- **Well-Standardized:** Currency, country, and categorical values standardized
- **Prepared for Analysis:** Derived numeric fields created from categorical data
- **No Duplicates:** Deduplication process applied successfully
- **Reasonable Outlier Handling:** Extreme salaries removed (outside $1K–$5M)

4.2 Limitations

- **Geographic Bias:** 70% USA-centric; limited international representation
- **Gender Skew:** 60% female respondents (likely Ask A Manager audience bias)
- **Self-Reported Data:** Respondents may under/over-report compensation
- **Missing Revenue Context:** No company size, profitability, or industry metrics
- **High Cardinality:** Job titles and industries have 600+/68+ unique values
- **Temporal Resolution:** Single-day snapshot (April 27, 2021 only)

4.3 Recommendations

- **NLP Processing:** Apply text mining to job titles and context fields
- **Industry Grouping:** Consolidate 68+ industries into 10–15 macro categories
- **Job Title Standardization:** Use classification models (e.g., O*NET mapping)
- **Currency Normalization:** Convert all salaries to USD for cross-country analysis
- **Demographic Analysis:** Be cautious with gender/race conclusions due to sample bias
- **Longitudinal Tracking:** Combine with future survey years for trend analysis